Lab programs

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DSA0504-QUERY PROCESSING

36. Scatter Plot for Three Different Groups Comparing Weights and Heights

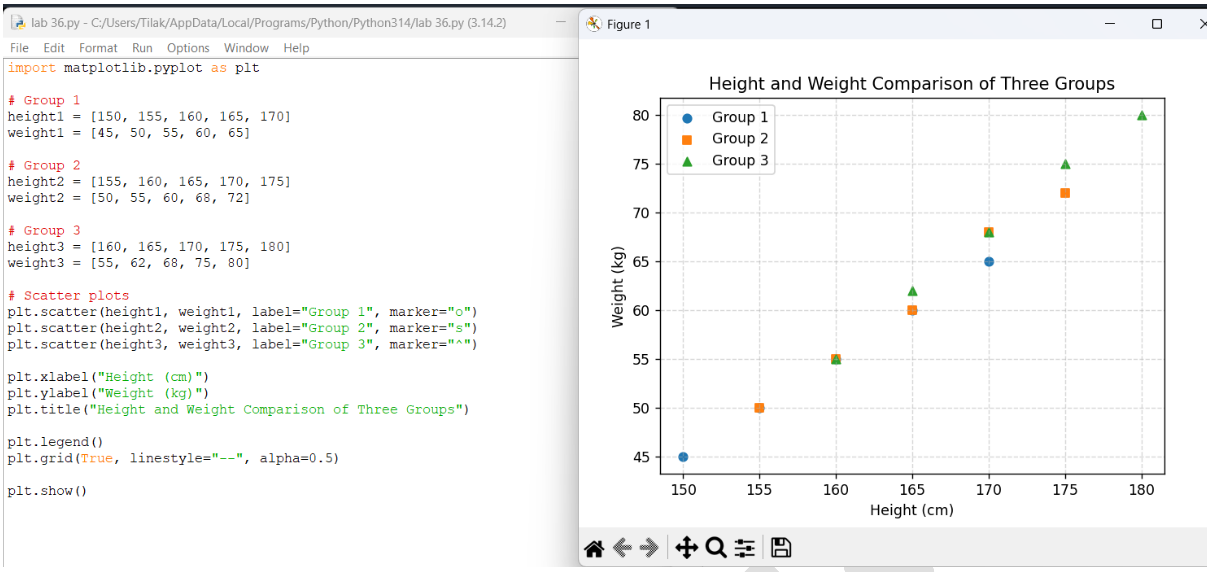
Aim

To draw a scatter plot for three different groups by comparing their weights and heights.

Code

```
import matplotlib.pyplot as plt

height1 = [150, 155, 160, 165, 170]

weight1 = [45, 50, 55, 60, 65]

height2 = [155, 160, 165, 170, 175]

weight2 = [50, 55, 60, 68, 72]

height3 = [160, 165, 170, 175, 180]

weight3 = [55, 62, 68, 75, 80]

plt.scatter(height1, weight1, label="Group 1", marker="o")

plt.scatter(height2, weight2, label="Group 2", marker="s")

plt.scatter(height3, weight3, label="Group 3", marker="^")


plt.xlabel("Height (cm)")

plt.ylabel("Weight (kg)")

plt.title("Height and Weight Comparison of Three Groups")


plt.legend()

plt.grid(True, linestyle="--", alpha=0.5)


plt.show()
```

Output



37. Create a DataFrame from a Dictionary

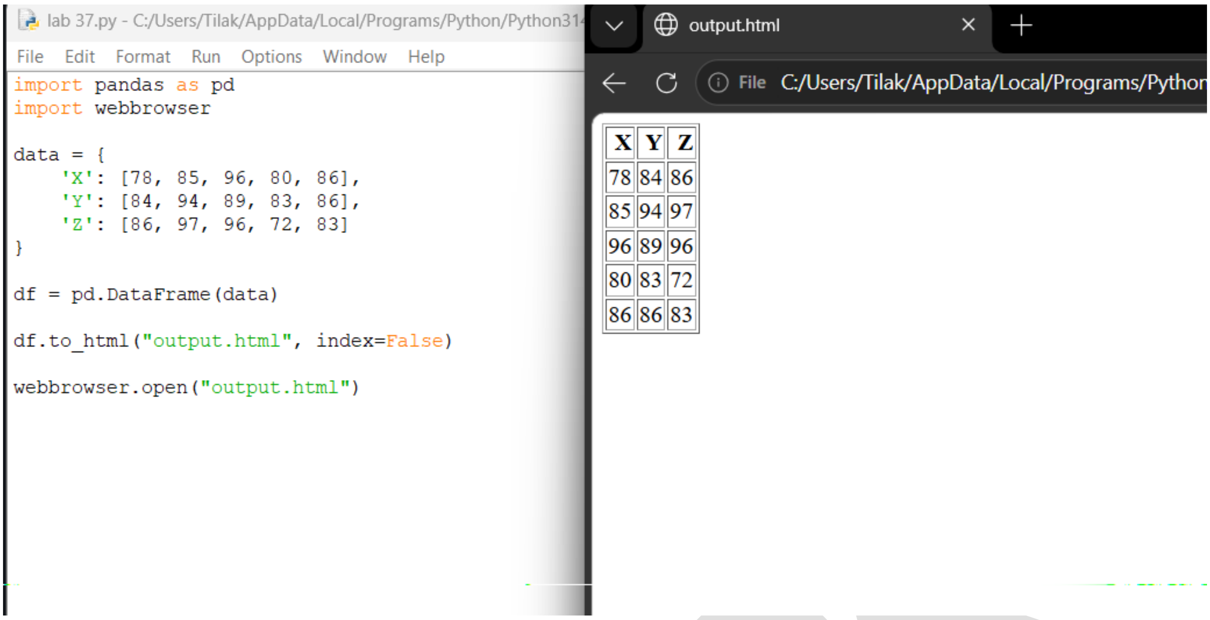
Aim

To create a Pandas DataFrame from a dictionary and display it.

Code

```
import pandas as pd

import webbrowser


data = {

    'X': [78, 85, 96, 80, 86],

    'Y': [84, 94, 89, 83, 86],

    'Z': [86, 97, 96, 72, 83]

}


df = pd.DataFrame(data)


df.to_html("output.html", index=False)


webbrowser.open("output.html")
```

Expected Output



38. Create a DataFrame from Dictionary with Index Labels

Aim

To create and display a Pandas DataFrame from a specified dictionary with custom index labels.

Code

```
import pandas as pd

import numpy as np

import webbrowser


exam_data = {

  'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',

         'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],


  'score': [12.5, 9, 16.5, np.nan, 9,

          20, 14.5, np.nan, 8, 19],


  'attempts': [1, 3, 2, 3, 2,

            3, 1, 1, 2, 1],
```

```
   'qualify': ['yes', 'no', 'yes', 'no', 'no',
               'yes', 'yes', 'no', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e',
          'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(exam_data, index=labels)

# Apply colors to the DataFrame
styled_df = df.style.set_properties(
   **{
      'background-color': '#EAF2F8',
      'color': 'black',
      'border': '1px solid black',
      'text-align': 'center'
   }
).set_table_styles([
   {
      'selector': 'th',
      'props': [
         ('background-color', '#2874A6'),
         ('color', 'white'),
         ('font-size', '16px'),
         ('border', '1px solid black')
      ]
   }
])
```

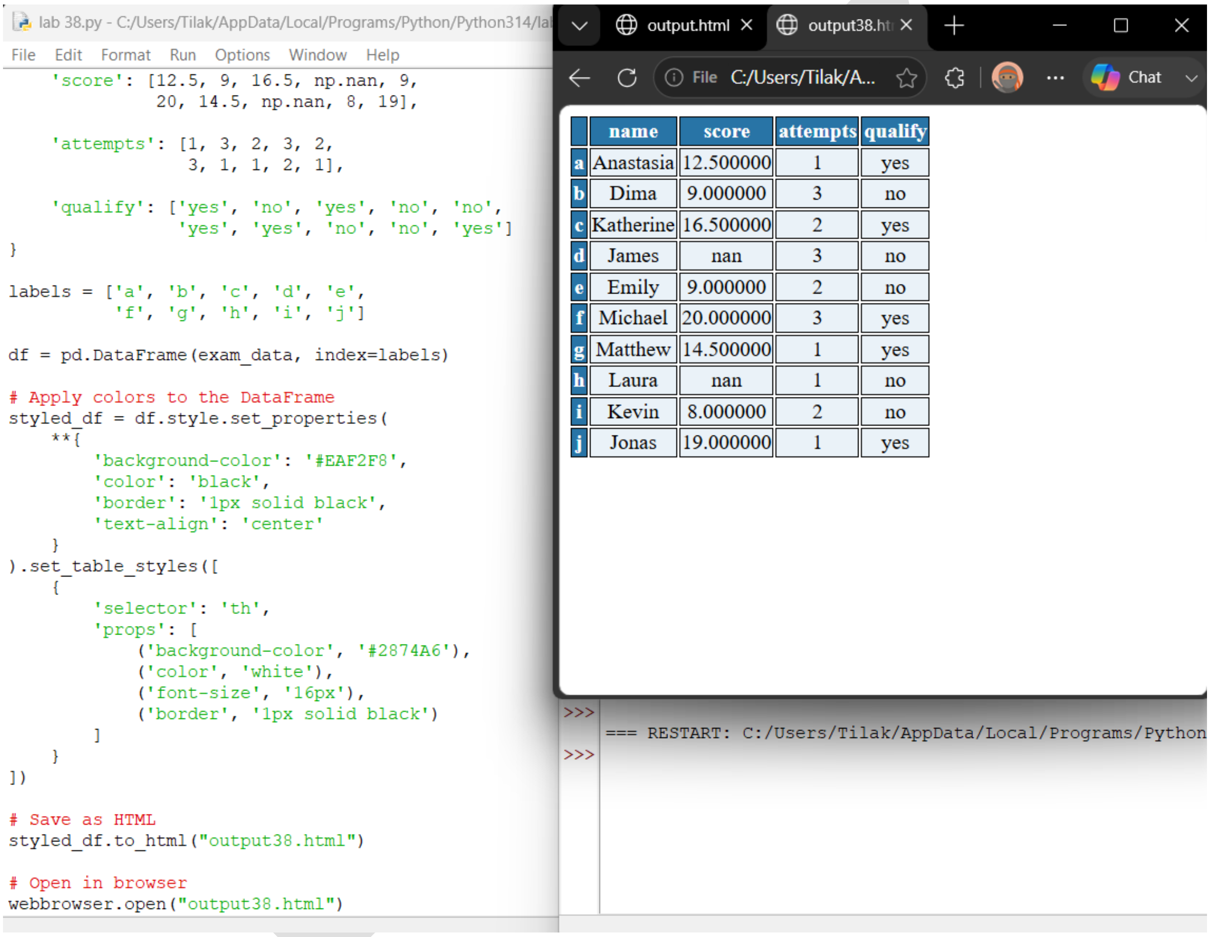
Save as HTML

styled_df.to_html("output38.html")

Open in browser

webbrowser.open("output38.html")

Output:



lab 38.py - C:/Users/Tilak/AppData/Local/Programs/Python/Python314/la

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```
    'score': [12.5, 9, 16.5, np.nan, 9,
              20, 14.5, np.nan, 8, 19],

    'attempts': [1, 3, 2, 3, 2,
                 3, 1, 1, 2, 1],

    'qualify': ['yes', 'no', 'yes', 'no', 'no',
                'yes', 'yes', 'no', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e',
          'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(exam_data, index=labels)

# Apply colors to the DataFrame
styled_df = df.style.set_properties(
    **{
        'background-color': '#EAF2F8',
        'color': 'black',
        'border': '1px solid black',
        'text-align': 'center'
    }
).set_table_styles([
    {
        'selector': 'th',
        'props': [
            ('background-color', '#2874A6'),
            ('color', 'white'),
            ('font-size', '16px'),
            ('border', '1px solid black')
        ]
    }
])

# Save as HTML
styled_df.to_html("output38.html")

# Open in browser
webbrowser.open("output38.html")
```

output.html | output38.ht

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	name	score	attempts	qualify
a	Anastasia	12.500000	1	yes
b	Dima	9.000000	3	no
c	Katherine	16.500000	2	yes
d	James	nan	3	no
e	Emily	9.000000	2	no
f	Michael	20.000000	3	yes
g	Matthew	14.500000	1	yes
h	Laura	nan	1	no
i	Kevin	8.000000	2	no
j	Jonas	19.000000	1	yes

```
>>>
=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python
>>>
```